MAZEN BAHGAT

DevOps Engineer

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PROFESSIONAL PROFILE

DevOps Engineer and AWS Certified Cloud Practitioner with an MSc in Computing (AI-Native Software Systems) from Queen's University and a DevOps & Infrastructure Engineering internship, automating the build, testing, deployment and monitoring of production services. Hands-on with CI/CD pipelines, Docker, Kubernetes, GitOps with ArgoCD, canary releases with Argo Rollouts, Terraform on AWS, and monitoring with Prometheus and Grafana. Writes Python and Bash automation, builds FastAPI back-end integrations, and resolves production issues through metrics, testing and root cause analysis. Brings an embedded engineering background in firmware-over-the-air updates, event-triggered camera systems and HIL testing, plus a peer-reviewed IEEE Access publication.

PROFESSIONAL EXPERIENCE

DevOps & Infrastructure Engineering Intern | Rivora Labs, Egypt *Aug 2026 – Sep 2026*

- Built back-end integrations in Python and FastAPI that pull Sentinel-2 satellite imagery (Copernicus APIs, with cloud-shadow masking) and hourly weather data into pipelines feeding a crop disease-risk model.
- Designed a secure FastAPI proxy between the dashboard and a self-hosted LLM (Ollama, OpenWebUI API), so model URLs, credentials and system-policy controls are never exposed to the client.
- Built the production field-health dashboard end-to-end: React and TypeScript front end, FastAPI and Uvicorn back end, Pydantic v2 validation, Argon2 authentication, SQLite storage and Vitest tests.
- Defined a PyTorch transfer-learning pipeline for crop-disease classification (EfficientNet, MobileNet, ResNet baselines), and designed the centre-pivot irrigation structure in Autodesk Inventor.

AUTOSAR CEA Intern | Appraid Tech (in partnership with Vector), Egypt *Sep 2024 – Oct 2024*

- Completed AUTOSAR Certified Engineer (CEA) training on automotive software architecture using PREEvision, DaVinci Developer, DaVinci Configurator, vVIRTUALtarget and CANoe.

Saher Radar Protection | Saudi Arabian technology company *Jun 2024 – Sep 2024*

- Built a multi-sensor radar-site security system (ATmega32) with event-triggered ESP32-CAM video capture and automated SMS and GPS alerts (SIM808) for real-time remote monitoring.

Professional Embedded Systems Intensive Training | NTI, Egypt *Mar 2024 – Jun 2024*

- Built an ADAS model car on STM32 with a firmware-over-the-air (FOTA) update mechanism over ESP8266 Wi-Fi, deploying firmware updates remotely without physical access.

Robotics & Electronics Instructor | Armstrong, Egypt *Sep 2023 – Mar 2024*

- Mentored 100+ students across multiple cohorts and guided 30+ concurrent groups, each completing a hands-on hardware project.

Embedded Controls Research Engineer | Arab Academy for Science and Technology, Egypt *Jun 2023 – Sep 2023*

- Designed a cascaded PI (position + velocity) controller for a linear servo mechanism with Wi-Fi telemetry, auto-generating production C code with MATLAB Embedded Coder.
- Published as: "Optimization of PI-Cascaded Controller's Parameters for Linear Servo Mechanism: A Comparative Study of Multiple Algorithms," IEEE Access, vol. 11, pp. 86377–86396, Aug 2023 (DOI: 10.1109/ACCESS.2023.3304333).

PROJECTS

HerHealthGPT: Multilingual Health Triage Platform | MSc Capstone, MLOps/Systems Lead, Queen's University *2026*

- Led MLOps and systems for a live multilingual (English, French, Arabic) women's-health triage assistant at herhealthgpt.org, mentored by the Deloitte Innovation Hub; co-authored the team's HerHealthEval benchmark paper on arXiv.
- Deployed on AWS with Terraform: Dockerized Next.js and FastAPI services on EC2 behind Caddy (Let's Encrypt HTTPS), RDS PostgreSQL in a private subnet, ECR, S3, Route 53, SES, Secrets Manager and SSM.
- Automated delivery with GitHub Actions: path-filtered CI, Git SHA-tagged images pushed to ECR, and SSM-run deploys that migrate the database, health-check the release and roll back automatically on failure; a CI gate blocks committed secrets.
- Served Qwen3.5-9B, a QLoRA adapter and Qwen3-Embedding-4B with vLLM on 4× NVIDIA A16 GPUs over Cloudflare tunnels with a self-healing watchdog, fixed a vLLM all-reduce fault that corrupted output, and set up CloudWatch logging with 7 SNS-routed alarms.

CI/CD Pipeline, GitOps & Observability Platform | Queen's University *Summer 2026*

- Automated build, test and release for a FastAPI ML inference API with a 7-step CI pipeline on Gitea Actions: Ruff, Mypy, pytest (100% coverage, 14 tests), multi-stage Docker build, Git SHA image tagging and Syft SBOM generation.
- Deployed updates to Kubernetes (k3d) through GitOps with ArgoCD (self-heal, prune) and Kustomize overlays, releasing via Argo Rollouts canaries in 20% steps with 30-second pauses.
- Built monitoring to catch production issues early: a Prometheus ServiceMonitor scraping /metrics every 15 seconds and a Grafana dashboard for request rate, 5xx errors, p50/p95/p99 latency and per-pod CPU and memory.

MLOps Pipeline: Canary Rollout, A/B Testing & Drift Monitoring | Queen's University *Summer 2026*

- Automated safe releases with an Argo Rollouts canary (20%→50%→80%→100%) gated by a Prometheus AnalysisRun on drift score < 0.5, promoting the new version only after 3 consecutive passes.
- Pushed drift metrics to Prometheus Pushgateway and provisioned a Grafana drift dashboard via ConfigMap; built a FastAPI A/B router (80/20 split) with Evidently AI drift monitoring across two model versions.

- Tracked experiments and model versions in MLflow (champion/challenger aliases); 4-bit GPTQ quantization cut inference latency 3.6x and model size 65%.

Cyber SOC Chatbot: Cloud-Based AI Security Triage Assistant | Queen's University *May 2026*

- Provisioned AWS infrastructure with Terraform (VPC, public and private subnets, NAT Gateway, security groups, IAM, S3, EMR), reproducible from a single terraform apply.
- Deployed the model on EC2 behind the Ollama REST API and Dockerized OpenWebUI with systemd auto-restart, least-privilege IAM, private-subnet isolation, IP-restricted SSH and encrypted EBS storage.
- Processed 114M+ security records with PySpark on AWS EMR and fine-tuned TinyLlama-1.1B with QLoRA, cutting training loss 58% in 41 minutes on a single GPU.

TestSeed: AI-Powered MongoDB Seed Data Generator | Queen's University *Jun 2026*

- Performed root cause analysis on AI generation failures (missing required fields, count mismatches, invalid ObjectId types) and fixed them with stronger prompt constraints and a tightened server-side validator.
- Used Spec Kit to turn acceptance criteria into a structured specification, breaking a 5-user-story epic into 63 dependency-ordered tasks, then completed all 63 with AI coding agents in 12 minutes 21 seconds.

Guarded RAG Tutor: Safe Retrieval-Augmented Generation System | Queen's University *2026*

- Built a RAG pipeline with LangChain, Sentence Transformers and ChromaDB, with OpenAI moderation and a Gemini 2.5 Flash scope classifier blocking 100% of off-topic and harmful test queries.

Portfolio: https://drive.google.com/file/d/1_McSX5Jai0DLpJoLd1LbKaMZ4IQe35xq/view?usp=sharing

EDUCATION

Master of Science in Computing (AI-Native Software Systems) | Queen's University, Canada *Sep 2025 – Sep 2026*

Delivered through the Digilians Initiative, including the 4-month NTI DevOps Track and 100+ hours of leadership and communication training.

Relevant Courses: DevOps & MLOps, Cloud Computing, Software Engineering with AI, AI & Machine Learning, Generative AI, Advanced Data Analytics

B.Eng. Mechatronics Engineering | Arab Academy for Science and Technology, Egypt *Oct 2018 – Jul 2023*

GPA: 3.5 / 4.0

TECHNICAL SKILLS

Cloud / DevOps: Docker, Docker Compose, Kubernetes, k3d, Helm, ArgoCD, Argo Rollouts, Kustomize, GitOps, CI/CD pipelines, GitHub Actions, Gitea Actions, Jenkins, Terraform, CloudFormation, Ansible, Infrastructure as Code, deployment automation, Caddy, Cloudflare Tunnel, Linux (RHEL)
Monitoring & Troubleshooting: Prometheus, Grafana, Prometheus Pushgateway, AWS CloudWatch, CloudTrail, Evidently AI, root cause analysis, production troubleshooting
Scripting & Programming: Python, Bash/Shell, YAML/JSON, SQL, TypeScript, JavaScript, C, C++, Embedded C, Java
AWS Services: EC2, S3, RDS, ECR, Lambda, API Gateway, DynamoDB, VPC, IAM, ECS, EKS, CloudWatch, CloudTrail, SNS, SES, Route 53, Secrets Manager, Systems Manager (SSM), EBS, EMR
Back-End & Integration: FastAPI, Uvicorn, Pydantic v2, REST APIs, React 18, Next.js, Vite, TanStack Query
Testing & Quality: pytest, Vitest, Ruff, Mypy, SBOM generation (Syft), HIL/SIL testing, JIRA (Agile/Scrum)
MLOps & AI Deployment: MLflow, vLLM, Ollama, PyTorch, HuggingFace Transformers, QLoRA/PEFT, GPTQ quantization, GGUF, ONNX Runtime, RAG pipelines (LangChain, FAISS)
Databases / Tools: PostgreSQL, MongoDB, SQLite, Git, GitHub, Jupyter, VS Code, Spec Kit, Cursor, Codex, Claude Code
Embedded & Modelling: AUTOSAR, RTOS, AVR/ARM/PIC/ESP MCUs, UART/SPI/I2C, CAN/LIN/FlexRay, MATLAB/Simulink, Embedded Coder, Autodesk Inventor

CERTIFICATIONS

- AWS Certified Cloud Practitioner - Amazon Web Services - Oct 2026
- Git Basics for DevOps - KodeKloud - Dec 2025
- DevOps Prerequisite Course - KodeKloud - Dec 2025
- Docker for Beginners with Hands-on Labs - KodeKloud - Nov 2025
- Docker Basics for DevOps - KodeKloud - Nov 2025
- Fundamentals of Red Hat Enterprise Linux - Red Hat - Nov 2025
- Introduction to Git and GitHub - Google - Oct 2025
- AUTOSAR CEA - Vector & Appraid Tech - Sep 2024
- Testing Foundations - Egypt FWD - Dec 2022
- Fundamentals of Deep Learning - NVIDIA - Jul 2022

VOLUNTARY & LEADERSHIP EXPERIENCE

Enactus Vice President | Arab Academy for Science and Technology *Jun 2021 – Oct 2022*

Student Union Vice President | Arab Academy for Science and Technology *Oct 2020 – Jun 2021*

LANGUAGES

Arabic (Native) - English (Fluent) - French (Beginner) - German (Beginner)